

# "Break the Mask Barrier": An AU-based Rehabilitation Training System for Parkinson's Hypomimia

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## Abstract

Hypomimia, commonly known as the "masked face," is a non-motor symptom of Parkinson's disease characterized by delayed facial movements and expressions, as well as difficulties in articulation and emotion. Currently, the primary method for detecting hypomimia relies on subjective evaluations by neurologists, and conventional rehabilitation approaches predominantly depend on verbal prompts from physicians. There remains a lack of accessible, patient-friendly, and scientifically rigorous assistive tools for hypomimia treatment. We aim to develop a digital therapy system based on Action Units (AUs) for the rehabilitation of Parkinson's hypomimia. This system enhances engagement through rehabilitation exercises set to Chinese opera music while also fostering initiative by integrating digital therapy with traditional facial training methods. A pilot study involving seven patients with hypomimia

symptoms and ten physicians was conducted. The results indicated a positive impact on participants' self-efficacy, validating the system's applicability in therapeutic settings and highlighting its potential value in clinical applications.

## CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

## Keywords

Parkinson's Disease, Digital Therapy, Rehabilitation Training, Masked Face, Hypomimia

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## 1 Introduction

Parkinson's disease (PD) is a chronic neurological disorder that predominantly affects individuals aged 60 years and older, necessitating long-term treatment and disease management [29]. The number of PD cases is rising rapidly in China, likely bringing a

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huge burden to the country's economy and healthcare system [33]. This necessitates research into more accessible, patient-friendly treatment options. Hypomimia, commonly known as the “masked face,” is a non-motor symptom of PD, characterized by diminished facial expressiveness and reduced movement [8, 41]. This condition can lead to complications such as dysarthria and is often associated with depression, emotional despondency, and cognitive impairment. Masked faces have become a significant barrier to the quality of life for people with PD.

Existing metrics, including the Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) [19], facilitate the assessment of Parkinson's hypomimia severity [20, 46, 47]. However, traditional evaluations rely heavily on in-person consultations with neurologists posing challenges due to time and resource constraints, which may prevent timely disease detection and intervention [8]. The integration of artificial intelligence (AI) techniques such as machine learning presents an opportunity for more efficient hypomimia detection, thereby enabling noninvasive monitoring and earlier intervention [2, 28, 32, 53]. Action Units (AUs) are the basic units of facial movement, providing a quantitative measure of facial expressions [35]. While previous studies have achieved high accuracy in recognizing facial AUs by utilizing AI models [17, 34, 39], gaps remain in the clinical application of these quantifiable measures for diagnosis and rehabilitation. Thus, developing objective assessment tools for facial rehabilitation of hypomimia and exploring AI integration into therapeutic training is a worthy area of study.

Targeted facial and tongue muscle training can effectively alleviate facial stiffness in PD patients. Patients' active engagement in the training is critical. Traditional facial exercises can be inconsistent due to varying conditions and severity of facial symptoms [30, 31]. Compared to this, digital therapies offer a more flexible and accessible alternative, potentially improving rehabilitation outcomes and quality of life. Moreover, these therapies can alleviate psychological burdens, fostering patients' engagement. Technologies such as online platforms provide convenient access to rehabilitation resources. Although some digital treatments for facial disorders, such as enabling users to match their expressions to 3D models [45] and using a mobile augmented reality mirror therapy system [7], comprehensive research on widely applicable digital therapeutic tools for hypomimia remains limited, especially in resource-constrained settings.

Enhancing self-efficacy in PD patients is crucial, as it empowers them to regulate their motivation and behavior, counteracting the adverse effects of emotional distress [21]. Self-determination theory posits that user motivation is influenced by autonomy, relevance, and competence [14]. Digital interventions can foster self-efficacy by encouraging positive behaviors and increasing confidence in symptom management. Traditional rehabilitation may enhance relevance and autonomy. Therefore, this study aims to allow users to interact through a familiar, effortless system and assess its potential as a valuable complement to existing research.

To break the mask barrier, we introduced an interactive system for detecting and rehabilitating facial expressions through the recognition of AUs. The detection process involves data acquisition and preprocessing, feature extraction and fusion of facial AUs, and classification of hypomimia, effectively capturing the reduced

facial movements associated with PD. Based on a formative study of current hypomimia rehabilitation clinical applications, we presented five key findings for system implementation. By combining traditional rehabilitation methods with digital therapy, we incorporated Chinese opera music with targeted exercises for various facial muscles. This approach quantifies facial movements, facilitating accurate assessments and enabling tailored rehabilitation programs. The contributions of this paper are: 1) integration of AI technologies with conventional detection and rehabilitation methods, providing seamless interaction and real-time feedback for patients with hypomimia; 2) development of an AU-based digital therapy system for detecting and rehabilitating hypomimia symptoms; 3) conducting a user study involving seven patients and ten physicians in a hospital setting to evaluate the clinical value of the system and its potential to enhance user engagement and initiative.

## 2 Related Work

### 2.1 Clinical Facial Detection and Rehabilitation for Parkinson's Disease

Various rating scales have been developed to assess the severity of Parkinson's disease (PD), including the Hoehn and Yahr staging scale [24], Montreal Cognitive Assessment [43], and MDS-UPDRS [43]. However, their application is often limited by the neurologist's experience, subjective assessment, and constraints such as time and distance. Although studies have utilized surface electromyography (EMG) to assess facial expressions and muscle activity [44], this method is time-consuming and requires specialized equipment. Therefore, there remains a need for a non-invasive detection method for patients.

Manual therapies have shown efficacy in improving facial motor function in patients [11, 30]. Chen et al. [11] reported positive outcomes from a passive massage rehabilitation exercise for patients with hypomimia. Li et al. [56] proved through experiments that the facial training method exercises the oral, facial, eyebrows, and eye muscles, which effected on improving facial movement problems significantly. Rehabilitation exercises were also demonstrated to effectively manage facial paralysis compared to traditional medications and acupuncture [25, 31]. Despite these advancements, clinical rehabilitation methods often lack real-time feedback and quantitative assessments, and limited healthcare resources hinder patients' access to professional guidance for self-assessing and training their hypomimia symptoms.

### 2.2 AI Technologies for Parkinson's Hypomimia

The potential of artificial intelligence (AI) in healthcare continues to expand, with deep learning enabling human-like performance across various clinical domains [9, 10, 22, 49, 51]. AI technologies have been applied in clinical treatment, medical research, and decision support for healthcare professionals. They offer a convenient diagnostic avenue, facilitating early intervention. Bandini et al. [5] extracted 49 facial key points and 20 features for hypomimia detection, while Grammat et al. [20] used FaceAPI to construct a linear regression model. AI-driven detection alleviates the mobility

challenges faced by PD patients, enabling remote diagnosis and reducing the barriers to timely assessment.

The Facial Action Coding System (FACS) [17] provides a framework for precisely characterizing facial expressions through Facial Action Units (AUs). Each AU represents specific facial muscle contractions or relaxations, allowing for detailed analyses of emotional and behavioral states. Liu et al. [39] introduced a deep learning framework using Graph Convolutional Networks (GCN) for AU detection. Li et al. [34] enhanced feature representation in facial regions through the AUs Semantic Relationship Embedding Representation Learning (SRERL) framework. While AI models have advanced AU recognition with high accuracy, significant gaps remain in translating these advancements into clinical practice for rehabilitation. Developing objective assessment tools and integrating AI technologies into rehabilitation strategies is essential for enhancing clinical engagement and even outcomes.

### 2.3 Digital Therapy for Parkinson's Hypomimia

Researchers have explored digital therapies for facial disorders, and some focused on interventions for early screening and rehabilitation [4, 16, 52]. Park et al. [45] developed a facial expression training system using bilinear shape models, enabling users to match their expressions with 3D models. Barrios et al. [7] introduced FaraPy, a mobile augmented reality mirror therapy system for facial paralysis. Liu et al. [38] designed a mobile game to analyze patient oral motor behaviors and track disease progression. However, comprehensive research on widely applicable digital therapeutic tools for hypomimia remains limited, and existing implementation strategies and evaluation criteria are not yet optimal.

Hypomimia frequently causes dysarthria, affecting 89% of patients and worsening with disease progression [13]. Cognitive impairment and mood disorders, particularly depression, further burden patients [1, 26, 36, 37, 48]. Music therapy can stimulate emotional expression and reduce stress, making it effective for addressing speech disorders. Saenz et al. [50] proved that can help to exercise the muscles and improve the state of facial stiffness through singing, while techniques like LSVT-LOUD [18] improve vocal symptoms. Visual and auditory stimuli were used to effectively exercise the oral and facial muscles, such as playing music with a strong rhythm and encouraging patients to shout slogans [3]. In addition, performing arts interventions have been shown to enhance the quality of life for PD patients [6]. Therefore, a personalized approach for PD patients should combine digital therapy techniques with conventional medicine and embed AI technology into facial movement analysis and real-time feedback.

### 3 Formative Study

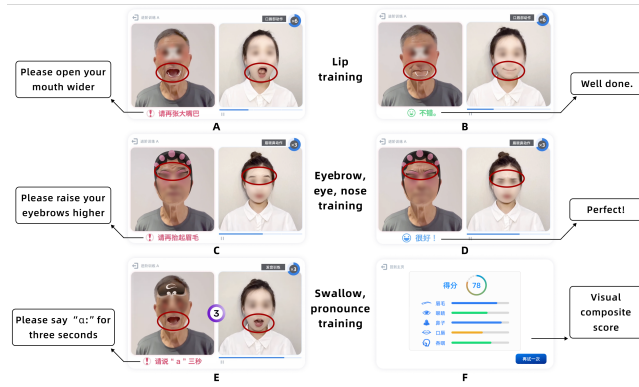
This study aims to explore the potential of digital therapies in reducing the learning burden of rehabilitation and increasing patient engagement and initiative. We conducted a formative study to gather insights from physicians specializing in Parkinson's disease on issues related to hypomimia and patients' needs for a digital therapy system. We collaborated with four neurologists and two rehabilitation physicians at a hospital in China, each with an average of over ten years of experience in their respective fields.

Semi-structured interviews were conducted to explore the practical applications of hypomimia therapy in rehabilitation.

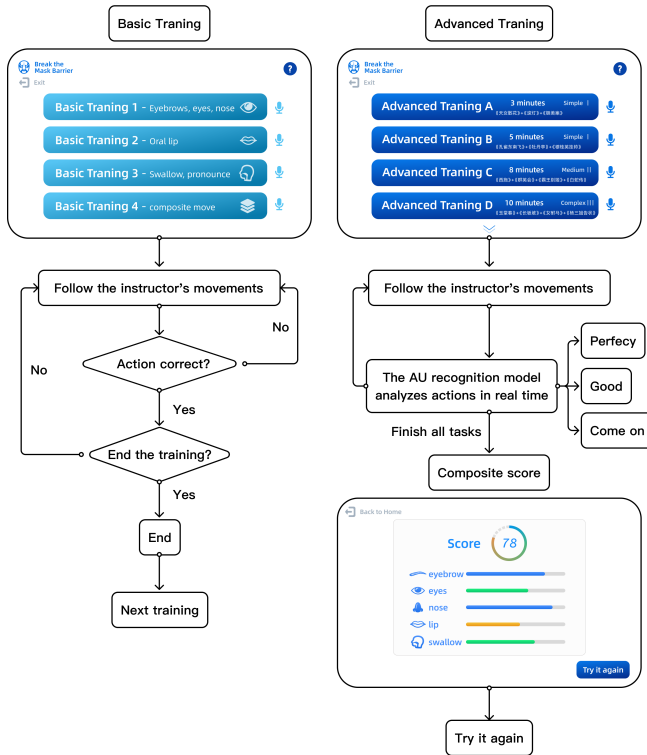
Here are our key findings: (1) **Cognitive Challenges in Patients with PD.** Hypomimia diagnosis heavily depends on neurologists' subjective expertise and requires substantial time and effort from patients in clinical settings. Cognitive deficits in attention and processing speed further hinder patients' multitasking abilities. Existing clinical digital therapies, such as eye-tracking games, neglect the complexity of facial movements. This underscores the need for a clear, intuitive user interface and a rational rehabilitation program that quantifies both the process and outcomes. (2) **Emotional Challenges.** Patients with PD commonly experience emotional disturbances like depression and anxiety. Impaired facial expression reduces their willingness to participate in exercises. Emotional acceptance of the disease and timely treatment are crucial for managing stress. Approaches that enhance daily living and foster appreciation for patients' capabilities, along with relaxation activities, can be beneficial. Thus, we must activate patient motivation through enjoyable rehabilitation experiences to elevate mood. (3) **Lack of Awareness in Active Rehabilitation.** The gradual onset of hypomimia symptoms can delay detection by both patients and healthcare providers. Active engagement in rehabilitation is essential, requiring sustained commitment even during symptom improvement. While recovery often relies on passive interventions, active treatment benefits from one-on-one guidance, sensory input, and immediate feedback. However, high costs and a shortage of therapists limit the use of active methods. Thus, our goal is to establish an accessible digital system that provides targeted guidance and fosters patients' autonomy through active contents. (4) **Continuous Monitoring of Symptoms.** Hypomimia severity varies with disease progression, with early symptoms often mild but intensifying over time. Self-monitoring is crucial for effective management and necessitates regular medical reviews. So our goals include managing data for treatment recommendations and providing understandable and visual feedback on rehabilitation. (5) **Categorization of Movement and Training Difficulty.** The variability of symptoms in Parkinson's hypomimia necessitates tailored rehabilitation strategies aligned with disease stages. Effective rehabilitation must address all facial regions to restore muscle flexibility and motor control, including throat and tongue functions crucial for articulation and swallowing. We aim to create a comprehensive training method that categorizes various facial actions and task difficulties.

### 4 System Design and Implementation

Based on previous research and our findings, we propose three design goals: (1) developing a professional and quantitative digital training program grounded in principles of medicine and rehabilitation, (2) creating interactive interfaces that provide timely, objective feedback for patients with hypomimia symptoms, and (3) ensuring continuous patient involvement and improving exercise initiative. We utilized the Vue framework and hosted the application on Python's built-in server. This system enables users to access it on laptops via web browsers in various environments, including hospitals, homes, and rehabilitation centers. It incorporates a facial recognition API.



**Figure 1: The interface during training: (A, C) prompt details of the facial movements to be adjusted; (B, D) feedback on how close the movement is to the standard; (E) the pronunciation training: singing a single tone aloud: “a;,” “ai,” and “i;.”; (F) visual evaluation of different parts of the face.**



**Figure 2: Procedure of basic training and advanced training. The basic training involves specific facial exercises with real-time feedback, prompting users to repeat movements until correct. The advanced training follows the appropriate duration and rhythm of Chinese opera, gives the user three levels of positive feedback, and finally scores the training.**

Building on research into facial training method for hypomimia rehabilitation in PD [11, 56], our system incorporates clinical facial

training that quantifies facial movement using data of AUs. We developed an interactive training system that integrates exercises for various facial regions with AU-based feedback, empowering users to actively engage in their recovery. To recognize AUs, we utilized the OpenGraphAU model [40], which identifies the movements of AUs associated with training and quantifies their activation levels. This model enables precise tracking of changes in facial expressions over time, as illustrated in Figure 1. The model first identified faces in the incoming video stream, selecting the one with the highest confidence for real-time analysis of the user’s facial expressions. Facial recognition showed good accuracy in healthy individuals.

We invited an acting teacher to record all the movements based on clinically proven facial training methods and edited them into videos. The system features two training modes: (1) **Basic Training**. Focuses on distinct facial regions, following a cycle of instruction, exercise, and feedback. The user selects a facial area for basic training, watches the video demonstration, and makes corresponding facial movements to get improved or complete feedback. (2) **Advanced Training**. The practice of integrated movements allows the user to choose the training content according to the duration and different music. The AU recognition model compares the user’s actions to a neutral baseline, analyzes the accuracy of the actions, and provides feedback categorized as “perfect,” “well done,” or “come on.” Ultimately, it provides a visual interface presenting facial training data that facilitates comprehensive monitoring and supports informed treatment methods. Detailed overview see Figure 2 and Table 1.

**Table 1: The details of facial movement training and the corresponding AU codes**

| Facial area      | Eyebrow                                                                                                                                     | Eye and Nose                                                                                              | Lip and Swallow                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Training action  | Frown exercise, eyebrow raising exercise                                                                                                    | Hard blink, nose wrinkle                                                                                  | Smile and grin, swallow training on the lips                                                                                                                                                                |
| Training content | Practice frowning and raising eyebrows repeatedly. Instruct the patient to focus both eyebrows in the center and perform forehead surprise. | Continuous eye opening and closing, opening and closing eyes forcefully, and nose stretching upward.      | Consistent movements include pouting, smiling, grinning, and opening and closing the mouth. Guide the patient to move the corners of the mouth outward and upward while pouting and bulging the cheeks.     |
| AU coding        | AU1: Raise the inner eyebrows and widen the eyebrows.<br>AU4: Lower eyebrows and frown hard                                                 | AU5: Lift upper eyelids and stare hard<br>AU44: Forceful blinking and closing eyes<br>AU9: Nose wrinkling | AU12: Mouth Corner Stretching<br>AU13: Cheek Puffing<br>AU19: Tongue Movement<br>AU27: Mouth Widening<br>AU24: Lip Pursing<br>AU18: Lip Pouting<br>AU33: Cheek Inflation for 3 Seconds<br>AU28: Lip Suction |

We integrate traditional Chinese opera, which is very popular among the elderly in China, into the rehabilitation program to enhance the training experience. With a centuries-old history and cultural influence [54], optimizes training effectiveness by matching the training content with the cultural background and rhythmic style of Chinese opera. For instance, the lively “Rolling Lantern” is

ideal for lip movement practice, while "The Case of Chen Shimei" aids in pronunciation with its articulate style. To make the training more fun, we incorporated opera-themed face filters tailored to each opera's style. They activated during transitions between opera segments, create an interactive "face-changing" effect, making the process engaging (Table 2 and Figure 3).

**Table 2: Content compatibility between Chinese opera and training**

| Chinese Opera Name                    | Concept for Filter Design                                                                                                                                                                                                                         | Training Content                                                                           |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| "Rolling Lantern"                     | It is a classic piece in Sichuan opera clown performances, renowned alongside face-changing and fire-spitting. The design is based on the humorous and whimsical style of the performance, creating a delightful and joyful atmosphere for users. | Its comedic and exaggerated nature is suitable for training lip movements.                 |
| "Celestial Beauty Scattering Flowers" | The rhythm of the opera transitions from slow to fast, expressing the joy of the celestial maiden as she departs from her heavenly realm and rides the wind. This can also evoke a cheerful and joyful mood for users.                            | The lively tempo is suitable for training eyebrow, eye, and nose movements.                |
| "The Case of Chen Shimei"             | Its character Bao Zheng upholds justice with integrity and impartiality. Every word resonates powerfully, uplifting the audience. It features clear enunciation and a tendency to elongate notes.                                                 | It's suitable for users to practice the pronunciation of Chinese characters (single tone). |



Rolling Lantern  
滚灯



Celestial Beauty Scattering Flowers  
天女散花



The Case of Chen Shimei  
铡美案

**Figure 3: The design of Chinese opera filters in this system. The opera filters cover the user's face in a translucent form to accommodate the user's facial movement during rehabilitation training.**

## 5 User Study

### 5.1 Procedure

We conducted the user study at a public hospital in China with: 7 PD patients with hypomimia (age:  $M = 69.57$ , MDS-UPDRS hypomimia severity score:  $M = 2.29$ ) with their caregivers; and 10 physicians (neurology department = 7, rehabilitation department = 3). Information see Table 3. The study was conducted in a communication room within the neurology ward. Participants were briefed on the study's aims. After obtaining informed consent, patients underwent initial assessments, including facial expressions and depression tests. Researchers then guided them through posture calibration. Participants received training on the system lasting 20-30 minutes, including a 3-minute advanced training and basic training sessions, with physicians and caregivers present throughout. Finally, questionnaires and semi-structured interviews were conducted (scenarios see Figure 4).

**Table 3: Demographics of the participants with hypomimia**

| ID  | Gender | Age | PD duration (years) | Hypomimia Severity | Depression Severity |
|-----|--------|-----|---------------------|--------------------|---------------------|
| PD1 | Male   | 71  | 3                   | 3                  | 2                   |
| PD2 | Male   | 63  | 1                   | 1                  | 3                   |
| PD3 | Male   | 65  | 16                  | 2                  | 5                   |
| PD4 | Male   | 80  | 3                   | 3                  | 3                   |
| PD5 | Male   | 72  | 1                   | 0                  | 2                   |
| PD6 | Male   | 62  | 4                   | 2                  | 3                   |
| PD7 | Female | 74  | 3                   | 5                  | 4                   |



**Figure 4: (a) A participant is being interviewed by the researcher; (b) a participant is using the system with the researcher's assistance.**

### 5.2 Measures

(1) *Assessment of Engagement and Autonomy.* We observed and analyzed participants' duration of training tasks and satisfaction to understand their initiative and engagement. We evaluated autonomy using questionnaires to assess participants' level of efficacy in specific tasks. And comparing participants' emotional assessments before and after using the system to clarify changes in self-efficacy. (2) *Assessment of the System's Usability.* Participants' success in the assigned tasks was evaluated by tracking errors and their instances of seeking help. Participants were asked to rate the training contents and forms of digital therapy by interviews. We employed the SUS with a 5-point Likert scale including clinical feasibility questions in the physician's questionnaire, and interviewed physicians to assess the system's therapeutic approach, functionality, and potential impact on patients. (3) *Interactive Performance of Patients.* Focusing on user interface issues and rehabilitation impact, we analyzed the duration and types of training tasks, measuring accuracy using AU values categorized into three levels. Participants provided feedback through interviews.

## 6 Results

### 6.1 Engagement of Participants

All participants completed both rounds of rehabilitation training tasks, with an average duration of three minutes for basic and advanced training. While overall accuracy varied among individuals, nearly all participants understood the training content and participated attentively. In the lip and articulation training, participants exhibited lower mean accuracy scores, with some misunderstandings in lip movements. Most participants grasped the articulation

movements, which effectively blended with the background music of Chinese opera. The majority felt the system was beneficial for improving pronunciation. Feedback indicated that some tasks were challenging, one participant noted, *“Simple movements are easy to learn. Combinations of movements felt difficult to do, and slower would be better.”* In addition, PD3 with early-stage Parkinson’s disease, stated, *“I noticed that my voice was weaker than it used to be, which made me reluctant to sing. I need to exercise my face more and talk more, but I don’t know what to do.”* Similarly, PD7 highlighted the positive impact of the system on her mood and rehabilitation goals, saying, *“I find this format highly beneficial. Listening to music while doing facial exercises may improve my mood. These exercises helped me feel my symptoms improve.”*

User satisfaction is crucial for assessing participation willingness. One participant, whose hypomimia symptoms severely affected his speech, participated in the questionnaire with the help of his caregiver. A 5-point Likert scale was used to measure satisfaction, yielding mean scores with interaction method ( $M = 3.29$ ,  $SD = 0.88$ ), opera music ( $M = 3.57$ ,  $SD = 0.73$ ), and operat filters ( $M = 3.33$ ,  $SD = 0.94$ ). Most participants found the music engaging and encouraging for collaborative singing, suggesting that systematic digital therapy is both attractive and effective. However, satisfaction with Chinese opera music varied based on individual preferences. Some participants remarked, *“I don’t care if there are opera filters as long as the rehabilitation training can alleviate the symptoms of my masked face.”*

## 6.2 Initiative of Participants

Questionnaire results showed that participants found the system engaging ( $M = 4.43$ ,  $SD = 0.90$ ) and expressed a desire for frequent use ( $M = 4.29$ ,  $SD = 0.70$ ). Most participants had no prior experience with digital therapeutics, but six found the approach convenient and were eager to adopt it. Participants especially valued the system for home use. PD5 stated, *“I can engage in rehabilitation when I have nothing else to do at home.”* PD6 noted its convenience compared to traditional methods, *“The doctor advised me to practice eight trigrams boxing, but I never tried it because my whole body had no strength. However, I feel I can participate in exercise with this system.”* This openness to new methods indicates high self-efficacy, though PD3 found no difference between the two modes. Physicians revealed that this system’s immediate feedback can enhance facial movement learning. One physician remarked, *“Active training is more effective because it creates a closed loop, and this form is very helpful for participants in initiating and regulating the movement.”*

## 6.3 The Appropriateness of the System

The system provided generally accurate data feedback during real-time AU recognition analysis, although there may have been delays or missed detections. AU recognition was systematically biased in the PD patient population, but the three levels of feedback provided confidence-building and thus sustained participation in the training, and did not directly alert the patient to movement errors, see Figure 1. Usability feedback was collected from participants and physicians. Through interviews, participants rated ease of use ( $M = 3.50$ ,  $SD = 1.12$ ), ease of learning ( $M = 4.67$ ,  $SD = 0.47$ ), comfort ( $M = 4.00$ ,  $SD = 0.71$ ), ease of recall ( $M = 4.67$ ,  $SD = 0.47$ ), and

fun ( $M = 3.29$ ,  $SD = 0.88$ ). These results indicate that the system meets usability needs and provides an intuitive user experience. Six participants reported feeling facial relief after training, five noted improved articulation, and all felt more emotionally relaxed. PD7 commented, *“I feel curious about the system. Feeling in a good mood because it feels helpful. If my doctor asks me to train every day, then I will take it.”*

Physicians scored fault tolerance ( $M = 3.88$ ,  $SD = 0.78$ ), clinical suitability ( $M = 4.00$ ,  $SD = 0.87$ ), and effectiveness ( $M = 4.29$ ,  $SD = 0.70$ ). Neurologists noted the system’s smoothness and positive participant experience, with comments like, *“The first patient seemed to be having a good time.”* Moreover, physicians emphasized active engagement in recovery and highlighted the system’s potential to slow disease progression and promote patient compliance, *“There is a lack of specialized software like this, which is truly needed.”*, and *“Oral facial exercises are essential and should be widely adopted in PD patients and performed regularly and comprehensively.”* Feedback suggested including positive reinforcement and encouraging statements to boost motivation. One neurologist also recommended a progressive training approach, starting with simple exercises and advancing to complex tasks, such as engaging in conversations or expressive reading.

## 7 Discussion

### 7.1 Reasonable Training Time, Frequency, and Strategy

In this study, we gain insights into the timing and methodology of rehabilitation training. Our rehabilitation system can optimize treatment plans based on patient-specific conditions and feedback. AI algorithms enhance diagnostic consistency by analyzing facial movements, allowing for a more objective understanding of the condition and recovery. This quantitative approach avoids the subjectivity of traditional assessments and enables accurate monitoring of rehabilitation outcomes. One participant suggested shorter training sessions, stating, *“It’s best not to make each session too lengthy; sitting for extended periods isn’t beneficial for the waist.”* Rehabilitation physicians recommended a home training regimen for long periods, suggesting five sessions a week, once or twice a day for 20 minutes. A physician noted that training should start with localized exercises and progress to overall training, beginning with oral exercises and integrating coordinated movements across the face.

Individual differences can lead to data biases, necessitating adjustments for patients with varying severity levels. It’s important to diversify the training data to include features from different populations and to optimize the speed and accuracy of real-time processing. Compared to patients with less severe symptoms of Parkinson’s disease, those with more severe symptoms, which are often accompanied by poorer cognitive performance and lower health status. Therefore, early detection and intervention of hypomimia are crucial, as patients in the early stages generally retain better cognitive function, making them more responsive to treatment. This system is expected to promote a sense of continuous recovery in patients, which is critical to improving outcomes and quality of life. These findings highlight the potential of the system.

## 7.2 Further Exploration of Rehabilitation Content and Function

Patients with Parkinson's hypomimia may struggle with complex and ambiguous movements [27]. Single-modal interaction systems often face challenges in user engagement, while multi-modal dialogue interfaces should account for users' cognitive abilities to ensure that the interaction modalities are both intuitive and easily comprehensible. Providing visual and auditory feedback aids patients in understanding the required actions, such as utilizing images, videos, or audio [55]. Furthermore, the supervision and guidance of healthcare professionals are crucial for support during using the system, as they can offer necessary support and interventions [23]. However, challenges persist in voice guidance, especially with dialect variations, highlighting the need for explanations from caregivers like family members. It's imperative to raise awareness of digital therapies among caregivers, and enable physicians to tailor training plans to patient progress with their caregivers' assistance [15]. Observations of the study revealed that some patients misinterpreted instructor gestures; for instance, during tongue exercises, when instructors pointed fingers at the cheek held up by the tongue, they mimicked the pointing gesture instead of engaging their tongue. This indicates that for complex or ambiguous movements, adding voice explanations is essential to help users understand the exercise.

There's also a need for further intensification of pronunciation training. Participants expressed a strong desire and acknowledged its effectiveness in improving articulation. However, the current system only includes vocal exercises for the single sounds: "a:", "ai," and "i:," there's also a need for further development. Research on computer-assisted pronunciation training has explored various approaches to improve learner outcomes [12, 42]. So we can evaluate the pronunciation function, such as analyzing speech parameters, and developing targeted training modules such as speech training and tongue twisters.

## 7.3 The Application Strategy of Hypomimia Digital Therapy

Early detection through AI algorithms enables timely rehabilitation, significantly impacting hypomimia intervention. Visual feedback enhances patients' understanding and management of their disease status. Digital therapies facilitate remote rehabilitation, benefiting patients in remote areas or with limited mobility and addressing healthcare resource disparities. In the long term, this approach can aid individuals with facial expression disorders integrate rehabilitation theory and active practice. The system exemplifies the versatility of digital rehabilitation therapy, offering effective support to a broader patient population and potentially shifting healthcare models toward more patient-centered approaches.

In addition, our study focused only on the musical context of Chinese opera, which suggests that patients have a limited interest in this genre. Future studies should explore a wider range of music genres such as pop music to accommodate different patient preferences. Combining styles and rhythmic elements of different music can increase patients' engagement and motivation. While preferences did not significantly impact engagement, regional variations

in musical tastes also suggest the importance of personalized music choices to improve patient autonomy and motivation.

## 8 Conclusion and Future Work

In regions with limited access to quality medical resources, timely diagnosis and rehabilitation of people with Parkinson's disease pose major challenges. We introduce an AU-based digital therapy system designed for diagnosing and rehabilitating hypomimia. The system extracts AU features for digital detection and offers systematic facial muscle training with real-time feedback. A formative study informed our findings, and then we conducted a user study with seven patients and ten physicians. The results confirmed the system's usability and enhanced users' engagement and initiative. It empowered patients to actively participate in their rehabilitation, boosting their self-efficacy and quality of life.

The study's findings may not be broadly applicable to all PD patients due to the limited sample size. Long-term evaluations in rehabilitation settings, such as nursing homes, are crucial for assessing the effectiveness of the system. In future research, we will incorporate a control group receiving conventional clinical interventions and conduct a randomized controlled trial (RCT) to compare our system with traditional rehabilitation approaches, demonstrating its impact and validating the clinical efficacy of the system for this population. Anonymizing patient identities during image processing while retaining diagnostic features is essential, and strong encryption standards must protect stored and transmitted data. Future efforts should also focus on personalizing rehabilitation programs based on individual patient conditions and feedback, as well as developing a comprehensive facial dataset to enhance model accuracy and efficiency.

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